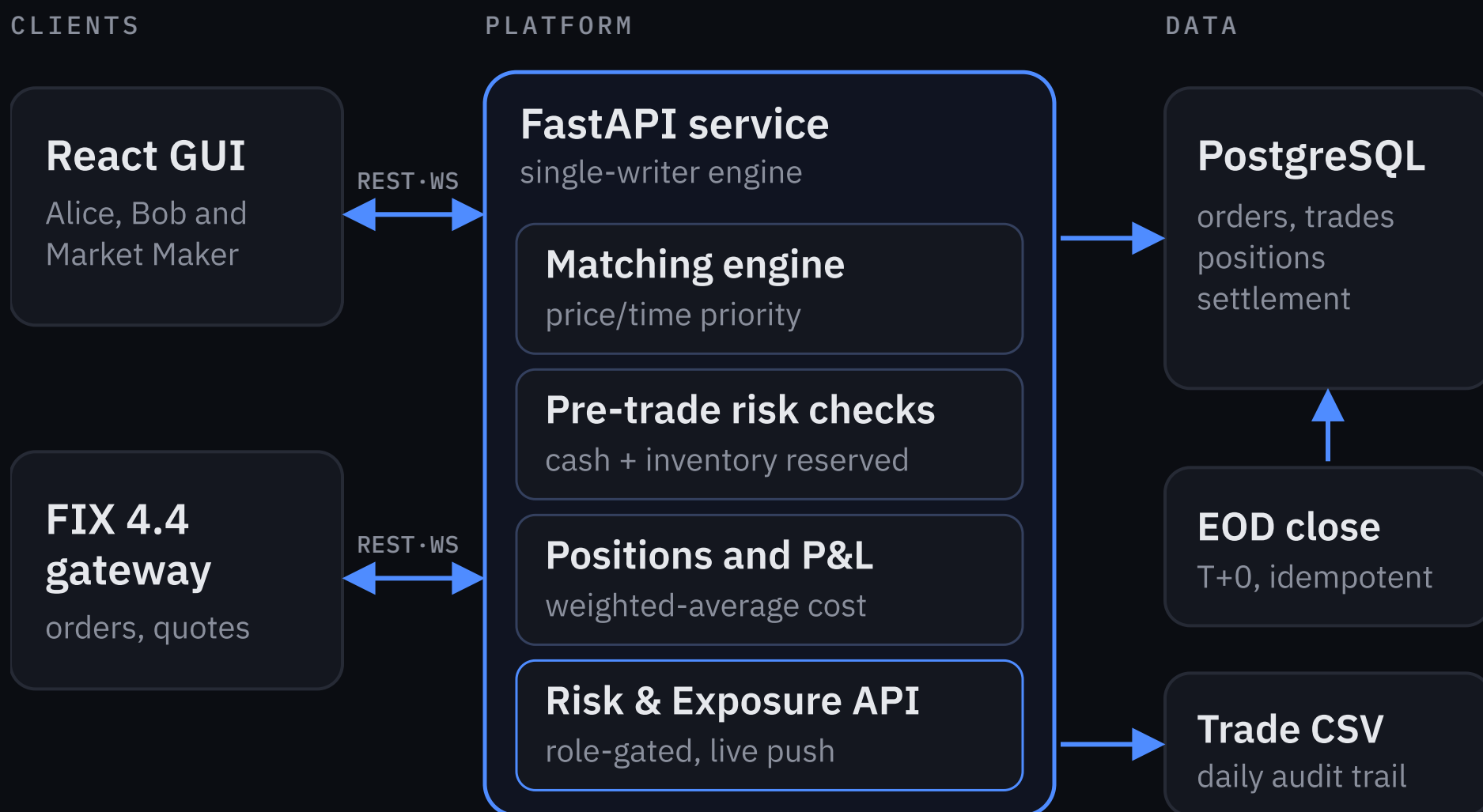


ARCHITECTURE

I built a trading platform and a risk engine, end to end.

Accounts trade each other through a real matching engine, and a risk cockpit watches every position.



AWS, defined in Terraform

validated, not deployed

ECS Fargate · RDS Postgres · ALB · S3 + CloudFront · EventBridge
44 resources, and no NAT gateway to keep the bill small

THE ENGINE

A real matching engine, not a price feed.

BTC/USD

ETH/USD

Price	Qty
61,350	15
61,200	12
61,100	8
61,050	5

spread 100.00

60,950	5
60,900	8.2
60,800	12
60,650	15
60,000	1.35

BUY

SELL

Type

LIMIT

Qty

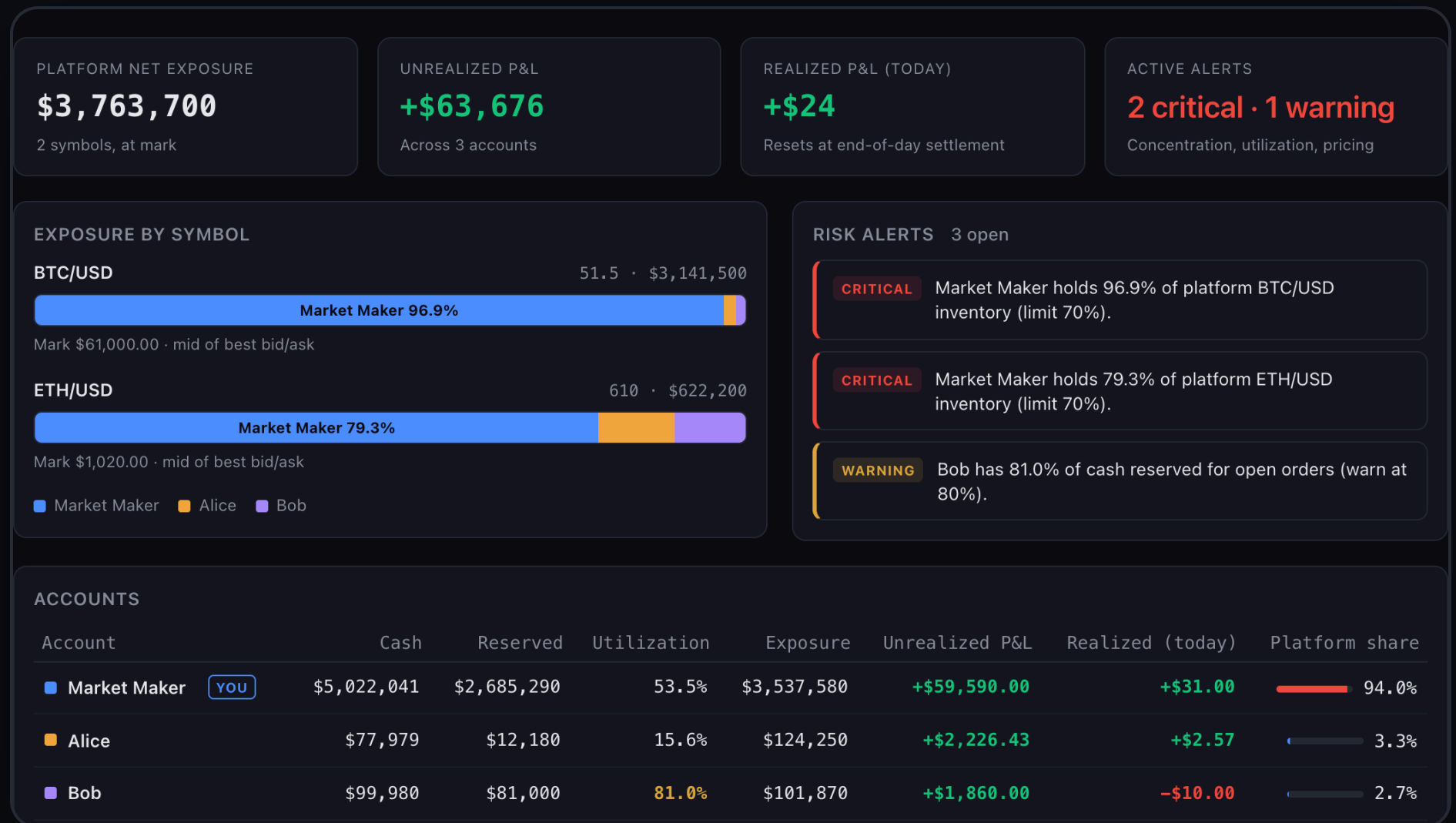
Price

BUY BTCUSD

- Price/time priority. **Limit and market** orders.
- Pre-trade checks reserve cash and inventory, so open orders can't over-commit an account.
- Invalid orders are **rejected, never queued**.

THE RISK ENGINE

Every account's exposure, live.



- Positions, P&L and concentration for **every account**, valued at mid, else last trade.
- No price yet? It shows as **unvalued**, never as zero.
- Pushed over WebSocket. Only the market maker can open it, **enforced on the server**.

ALERTING

It flags the problem before a person has to.

RISK ALERTS 3 open

CRITICAL Market Maker holds 96.9% of platform BTC/USD inventory (limit 70%).

CRITICAL Market Maker holds 79.3% of platform ETH/USD inventory (limit 70%).

WARNING Bob has 81.0% of cash reserved for open orders (warn at 80%).

CRITICAL One account holds over 70% of a symbol's inventory

WARNING Reserved cash is 80% or more of an account's balance

WARNING A held symbol has no price, so exposure is unvalued

Thresholds are config, not code. Alerts are computed live from current positions, so they clear on their own.

AUDIT TRAIL

Every fill and every reject, on the record.

Executions 4

Order ID	Trade ID	Symbol	Side	Qty	Time of Trade	Price	Status
21	—	BTCUSD	SELL	5	12:40:51	60,000	REJECT
19	5	BTCUSD	SELL	0.1	12:40:51	60,040	FILL
17	3	ETHUSD	BUY	10	12:40:51	1,001	FILL
15	1	BTCUSD	BUY	0.3	12:40:51	60,050	FILL

trade_logs/YYYYMMDD.csv · one trade, both counterparties

trade_id	account	side	order_price	fill_price	qty	realized_pnl
5	bob	BUY	60,040	60,040	0.1	0
5	alice	SELL	60,040	60,040	0.1	+2.5714

- Rejects **keep their reason**. Nothing is silently dropped.
- Both sides show the order price against the price executed, and P&L at that moment.
- **T+0 settlement**, with an idempotent end-of-day close.

RESULTS

What I'd bring to your desk.

31

automated tests passing, including
a FIX end-to-end

3_{ms}

median order ack, and 9 ms when
it fills

43_{ms}

from a fill to the risk view update

5.7_k

lines of Python, TypeScript and
Terraform

WHERE IT MAPS TO THE JOB

- Pre-trade risk controls and entitlements
- Position, P&L and settlement logic
- FIX connectivity and real-time data
- Cloud infrastructure as code, cost-aware

Open to roles in trading technology and risk.

Message me and I'll walk you through the demo.

A simulation, not production. The FIX gateway is a template and the AWS stack is validated, not deployed. Latencies measured on a laptop with a local database.